SETUP

Different devices. The same Python.

Choose where Python will run

Colab

No local Python installation. Save a copy and upload the course support files.

Local JupyterLab

Install Python 3.12 or a newer supported stable release, plus JupyterLab. Download the course package.

Keep the choice simple

Use the route that works on your device. No GPU, paid plan or API key is required.

Colab uses a cloud runtime; local JupyterLab runs on your computer.

Get a complete copy of the materials

1

Keep a backup

Keep the downloaded ZIP unchanged, then extract a working copy.

2

Get the package

Open the extracted root README before choosing a notebook.

3

Keep the folders

Keep learning_materials/, its assets/, programs/ and final-data/ together.

4

Choose the week

Open the week’s notebook and follow its first setup instructions.

Opening an .ipynb file alone does not bring its data or programs with it.

Windows: create a local environment

```
py -3 --version
py -3 -m venv .venv
$coursePy = ".\.venv\Scripts\python.exe"
& $coursePy -m pip install jupyterlab
& $coursePy -m jupyterlab --notebook-dir=learning_materials
```

1. Create .venv beside learning_materials/.
2. Use its Python for installation and launch.
3. Open the week’s notebook in JupyterLab.

Run these commands in PowerShell from the extracted repository root.

macOS or Linux: create a local environment

```
python3 --version
python3 -m venv .venv
.venv/bin/python -m pip install jupyterlab
.venv/bin/python -m jupyterlab --notebook-dir=learning_materials
```

1. Ubuntu is one Linux distribution.
2. Use .venv/bin/python for this project.
3. Keep system packages separate.

Run these commands in Terminal from the extracted repository root.

Colab: prepare your own working copy

1

Open the notebook

Use its Colab badge or upload the downloaded .ipynb.

2

Save your copy

Save in your own Drive or download your working notebook.

3

Prepare files

Follow the first setup cell to upload the course ZIP.

4

Check and run

Confirm the learning_materials/ working folder, then run from the top.

A Colab notebook link opens a notebook; prepare its support files separately.

Verify the interpreter, folder and calculation

```
01 import sys
02 from pathlib import Path
03
04 print(sys.version.split()[0])
05 print(sys.executable)
06 print(Path.cwd())
07 print(3 * 4)
08 assert Path("programs").is_dir()
```

1. Check Python3 and your working folder.
2. The last number should be 12.
3. programs/ should be available.

The interpreter and current directory explain many “it works on my machine” problems.

Save the notebook and the files you create

Notebook

Save your edits and explanations. Reopen your copy to confirm they remain.

Generated files

Keep scripts, data and databases that your exercise needs. Colab runtime files are temporary.

Clean rerun

Restart the kernel/runtime as instructed and run from the top using known input files.

A saved notebook is not a backup of every file in its runtime.

An error is information you can use

Name the environment

Your OS or Colab, Python version and the notebook week.

Show the evidence

Input, expected result, observed result and the relevant traceback.

Say what you tried

One attempted fix and its result. Search Issues first; post the successful follow-up.

Report a small example and the exact error, not only “it does not work”.

You are ready when these checks pass

Check	Evidence
Access	You can open the current week and support files.
Execution	A notebook cell runs and reports 12.
Files	The programs/ directory is available.
Persistence	Your saved notebook reopens with your edits.
Help	You can open an Issue with a small example.

Use SETUP.md for the full guide and Issues for shared setup questions.

Further reading in English

1. ## Python — virtual environments and direct interpreter paths

   https://docs.python.org/3/library/venv.html

2. ## Jupyter — installing JupyterLab

   https://jupyter.org/install

3. ## Google Colab — notebooks from GitHub

   https://github.com/googlecolab/colabtools/blob/main/notebooks/colab-github-demo.ipynb

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Read one relevant section, then bring an example and a question.